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Summer Populations of Northern Long-eared Bat in an Eastern Kentucky Forest Following Arrival of White-nose Syndrome

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ABSTRACT.—*Myotis* bats have experienced significant population losses due to white-nose syndrome (WNS) throughout large portions of their distributions in eastern North America. As closed-space foragers, these species comprise an important feeding guild within eastern forests. An understanding of where summer populations remain and how their ecology has changed following impact from WNS is needed to assess fully the recovery potential of *Myotis* bats. We used acoustic sampling, capture surveys, radiotelemetry and roost surveys from 2015 to 2018 to evaluate the status of northern long-eared bat (*Myotis septentrionalis*) in an eastern Kentucky forest following region-wide impacts from WNS to this species. Acoustic activity of *Myotis* remained unchanged over the 4 y of sampling, with activity of these bats greatest in mid- to late July. Northern long-eared bats represented 97% of the *Myotis* captured, indicating activity levels likely reflected patterns for this species. We located 18 roost trees of northern long-eared bats in five tree species, including both live and dead trees. All roost trees were on upper slopes within 100 m of ridge top roads. Maximum exit counts, 24 and 21, at two roosts occurred in late May and early June, suggesting these dates represented the summer maternity period of northern long-eared bats in this forest. Our results demonstrated that post-WNS populations of northern long-eared bats on Robinson Forest formed small colonies, day-roosted in trees near roads on forested ridge tops, were reproductively active, and maintained steady levels of activity across the 4 y of sampling. These data suggest that local populations of northern long-eared bats are surviving WNS and continue to persist during summer months in forests of the Appalachian Mountain region. We hypothesize presence of ridge top roads, often associated with forest logging operations, may be important habitat elements for increasing availability of preferred roosting habitat for summer populations of northern long-eared bats in actively managed forests.

INTRODUCTION

White-nose Syndrome (WNS) is a recently emerging disease of North American bats caused by the psychrophilic fungus, *Pseudogymnoascus destructans* (Blehert *et al.*, 2009; Frick *et al.*, 2016). Eastern North American populations of cave-hibernating species have been decimated by the fungus with mortality numbers in the millions. Presently, several *Myotis* species are known to be severely impacted by the fungus, including little brown bat (*Myotis lucifugus*), northern long-eared bat (*M. septentrionalis*), and Indiana bat (*M. sodalis*) (Thogmartin *et al.*, 2012, 2013; Vonhof *et al.*, 2015, 2016; US Department of the Interior, 2015; US Fish and Wildlife Service, 2019).

Studies are beginning to show that some populations of cave-hibernating bats that have survived the initial impact of the fungus are adopting alternative behaviors and hibernation strategies, including use of alternate roosts, increased grooming, reducing cluster size, and moving to warmer or cooler microclimates (Brownlee-Bouboulis and Reeder, 2013; Grieneisen *et al.*, 2015; Johnson *et al.*, 2016; Mayberry *et al.*, 2018; Webber and Willis, 2018). Further evidence also suggests that populations of some species are evolving resistance to the pathogen, including larger body mass associated with WNS survivors in several bat species across multiple cave systems (Frank *et al.*, 2014, 2019; Lacki *et al.*, 2015).

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Local populations of bat species in some infected areas are now beginning to stabilize or even increase in numbers slightly (Reichard *et al.*, 2014; Dobony and Johnson, 2018). Regardless, surviving populations of affected bats remain vulnerable, are poorly documented, and possess low reproductive rates that will still take decades to recover (Dobony and Johnson, 2018). Therefore, identifying forested habitats and geographic locations where survivors of WNS spend the summer months living and rearing young is vital information for use in recovery efforts of WNS-affected bats.

For this study, we investigated summer populations of northern long-eared bats in Robinson Forest, Kentucky. Robinson Forest is an experimental forest reserve, which historically maintained summer populations of northern long-eared bats prior to arrival of WNS. We used acoustic sampling, capture surveys, radiotelemetry, and roost surveys to evaluate the abundance, roosting patterns and reproductive potential of summer populations of northern long-eared bats from 2015 to 2018. Historically, several *Myotis* species were known to occur in this forest, with northern long-eared bats being common and easily captured (Krupa and Lacki, 2002). White-nose Syndrome reached Kentucky by the winter of 2010-11 and has since spread across the state, including cave systems within the Cumberland Plateau physiographic province, the landscape region of the study area (Kentucky Department of Fish and Wildlife Resources (KDFWR), unpubl. data). We predicted northern long-eared bats would still be present in the forest, but hypothesized that rates of detection by capture and acoustic surveys would be low, and that maternity colonies of this species would be small compared with observations prior to WNS elsewhere in Kentucky and surrounding regions.

STUDY AREA

The study area was located within the Cumberland Plateau physiographic province in eastern Kentucky. Eastern Kentucky has elevations ranging from 200–500 m (McGrain, 1983), with rugged terrain largely covered by mixed mesophytic forests (Braun, 1950). Eastern Kentucky possesses extensive sandstone cliffs, rock outcrops and a variety of caves formed from layers of both sandstone and limestone bedrock throughout the region (McGrain, 1983; Simpson and Florea, 2009). The climate of the region is temperate humid continental, with no distinct dry season and warm summers and cool winters. Annual precipitation averages 114 cm with a mean annual temperature of 13.9 C (Hill, 1976).

Robinson Forest (N 37.45023/ W83.19541) is an experimental forest, which lies within Breathitt, Knott and Perry counties, Kentucky, and is managed by the University of Kentucky and the KDFWR. The main block of the forest exceeds 4000 ha with outlying fragmented parcels comprising an additional 2000 ha. The forest is surrounded by strip-mined lands in various stages of ecosystem recovery and is isolated from other intact forests. Robinson Forest was nearly clear cut for the sale of lumber, with commercial harvesting of timber ending by 1922 (Krupa and Lacki, 2002). Robinson Forest has been subjected to many types of disturbance since then, including experimental harvesting treatments, wildfires, mining and invasion by exotic plants (Krupa and Lacki, 2002). The forest has a maintained road system, allowing access to more remote portions of the main block. Despite these impacts, Robinson Forest has developed into a mature, second-growth forested landscape with diverse plant and animal communities. Bottomlands are mesic and comprised of maple (*Acer*)-beech (*Fagus*)-poplar (*Liriodendron*) stands, with hemlock (*Tsuga*)-*Rhododendron* communities interspersed. Mid-slopes support oak (*Quercus*)-beech-maple forest. Forested habitats on ridge tops are comprised of oak-pines (*Pinus*) or oak-hickory (*Carya*) stands due to xeric, sandy soils.

MATERIALS AND METHODS

We surveyed acoustic calls of *Myotis* bats during the summers of 2015 to 2018 to index the relative abundance of these bats across season and years. In 2015, we sampled acoustic points during two sessions from 17 June to 16 September. During 2016, we sampled acoustic points during three sessions from 23 May to 11 September. In 2017, we sampled acoustic points during two sessions from 23 May to 20 July. Finally, in 2018, we sampled acoustic points during two sessions from 22 May to 13 July. We monitored activity using nine Song Meter 3 (SM3) units and SMU-1 microphones (Wildlife Acoustics, Maynard, MA). In 2015 and 2016, we housed SM3 units within pelican cases, with microphones placed within polyvinyl chloride pipe and tied to a tree at 1.5-m aboveground. Housing was intended to prevent damage from the elements and wildlife. The additional shielding created secondary harmonics, limiting the quality and resolution of call characteristics. Therefore, we removed the shielding for sampling in 2017 and 2018. During each session, acoustic sampling occurred for a minimum of three consecutive nights to account for random variation in nightly activity patterns. We collected data from sunset to sunrise each night of sampling.

We established six acoustic transects throughout the Laurel Ridge subunit of Robinson Forest (N 37.45023/ W83.19541) with transects selected based on randomly chosen distances and directions within the study area. Transects contained a ridge top, mid-slope, and riparian sampling point. We used two predominant slope directions, *i.e.*, north to south or east to west. The closest ridge top to a random point became the starting point of a transect. We placed the riparian point adjacent to the closest stream to the selected ridge top, with mid-slope points placed at an elevation halfway between the riparian and ridge top points. When possible, we located units in the vicinity of closed canopy roads, streams and canopy gaps. In 2018, an experimental harvest was implemented in Laurel Ridge that created shelterwood harvests and patch cut harvests (Arant, 2020). These changes overlapped the acoustic transects, such that in 2018, two transects fell within uncut forest, two were within a network of patch cuts, and two occurred within a shelterwood harvest. This led to slight adjustments in our placement of some acoustic units from the original transect layout due to loss from the harvest of trees originally used for mounting equipment and due to reorienting of some units to survey patch cut openings. We ensured that placement of units was at distances <5 m from the original point.

We analyzed acoustic data using Kaleidoscope v. 3.1.8 (Wildlife Acoustics, Maynard, MA; www.wildlifeacoustics.com). Kaleidoscope has been used for acoustic bat studies (Austin *et al.*, 2020) and is approved for acoustic surveys of bats by the U.S. Fish and Wildlife Service (<https://www.fws.gov/midwest/endangered/mammals/inba/surveys/inbaAcousticSoftware.html>). Kaleidoscope identified ultrasonic recordings as either bat calls or background noise. Data were compiled and organized using 'R' statistical software 3.5.0-Joy in Playing (R Development Core Team, 2013). R programming was used to sort and filter call data for identification and to separate out calls of *Myotis* bats comprised of ≥ 4 sequential pulses. We used the number of calls of *Myotis* bats as the measure of activity. We compared mean rates of *Myotis* bat activity (calls per detector-night) monthly and yearly using one-way analysis of variance (ANOVA; PROC GLM; SAS 9.4, SAS, Cary, NC, USA).

We captured bats in mist nets from 19 May to 20 August 2016, 9 May to 1 August 2017, and 23 May to 13 July 2018. We determined location of netting sites, in part, based upon results of previous acoustic surveys demonstrating activity of *Myotis* bats in the vicinity using Kaleidoscope (Arant, 2020). Nevertheless, we captured bats in Robinson Forest in a variety of habitats: clearings with human structures, over creeks, along forest roads, across post-harvest skidder trails, and in patch cut and shelterwood harvests. Net widths ranged in

length and varied from single nets (2.6 m) to double- and triple-stacked (5.3 m; 7.8 m) nets, depending on the location surveyed. Nets were raised using Avinet poles (Dryden, NY) as single highs, and as double and triple highs with the forest filter system (Bat Conservation and Management, Inc., Carlisle, PA).

We collected data on all bats captured, including mass (g), right forearm length (mm), reproductive condition (Racey, 2009), sex, age (Brunet-Rossinni and Wilkinson, 2009), height in net, and presence of ectoparasites. We determined age by shining a light through the joints of the finger bones. Adult bones are ossified and light does not pass through. Juvenile bones are not fully ossified and light passes between the bones in the finger joints. We assessed pregnancy in females by the presence of a fetus in an enlarged abdomen. We determined reproductive state of adult females as lactation, when a patch of hair around the mammary glands was absent. We determined reproductive status of males by examining for distension of descended epididymis.

We attached 0.27-g radio transmitters to 16 northern long-eared bats and tracked them to roost trees. We glued LB-2XT transmitters (Holohil Systems, Ltd., Ontario, Canada) between the shoulder blades of bats with surgical cement (Perma-Type Company, Inc., Plainville, CT). We tracked radio-tagged bats to day-roosts using three or five element yagi antennae (Wildlife Materials, Inc., Murphysboro, IL), combined with either Icom IC-R20 radio receivers (Icom America, Inc, Kirkland, WA), R-1000 receivers (Communication Specialists, Inc., Orange, CA), or TRX-2000 receivers (Wildlife Materials, Inc., Murphysboro, IL). We searched for bats each day until the transmitter was found dead or the bat could not be located for three consecutive days. In the latter instance, we only periodically checked for the signal beyond the 3 d limit.

We identified tree roosts to species and decay class (Maser *et al.*, 1979), and measured stem diameter (cm). We geolocated all roosts within 5-15 m with a Garmin GPSMAP 64 unit. We compared diameter size of tree roosts in May (prepregnancy) with those found in June and July (pregnancy and lactation) using a Student's *t*-test based on equal population variances (Daniel, 1974). We sampled available trees at randomly-chosen plots associated with each roost tree identified. We assigned random plots to either a ridge top or mid-slope position; the only landscape positions in the study site where we found northern long-eared bats roosting. We determined plot locations using a random compass orientation between 0 or 180 degrees and a random distance >50 m from a known roost tree. We measured all trees within a 10-m radius around each random sampling point. We collected data on species, decay class and presence of cavities for all stems with a diameter greater than 4 cm; the smallest diameter of tree used as a roost by northern long-eared bats in this study. We evaluated use of cavity trees by species with a Chi-square goodness-of-fit test, with less common species combined into a single category to ensure that no categorical grouping had an expected frequency <1 (Cochran, 1954). When bats were tracked to roost trees, we performed exit counts of bats (*i.e.* the number of bats emerging on a nightly basis) the first night after a new roost was discovered to assess population size of bats at that roost. Exit counts started 20 min before sunset and ended after 10 min had passed without a bat emerging from the roost. Personnel positioned themselves in an orientation that ensured the bats were silhouetted against the sky.

RESULTS

In total 84 nights of sampling yielded 708 detector-nights in the Laurel Ridge experimental harvesting area. Call rates of *Myotis* were low across the four years of sampling. Evaluation of *Myotis* activity by month demonstrated a peak in activity during mid-

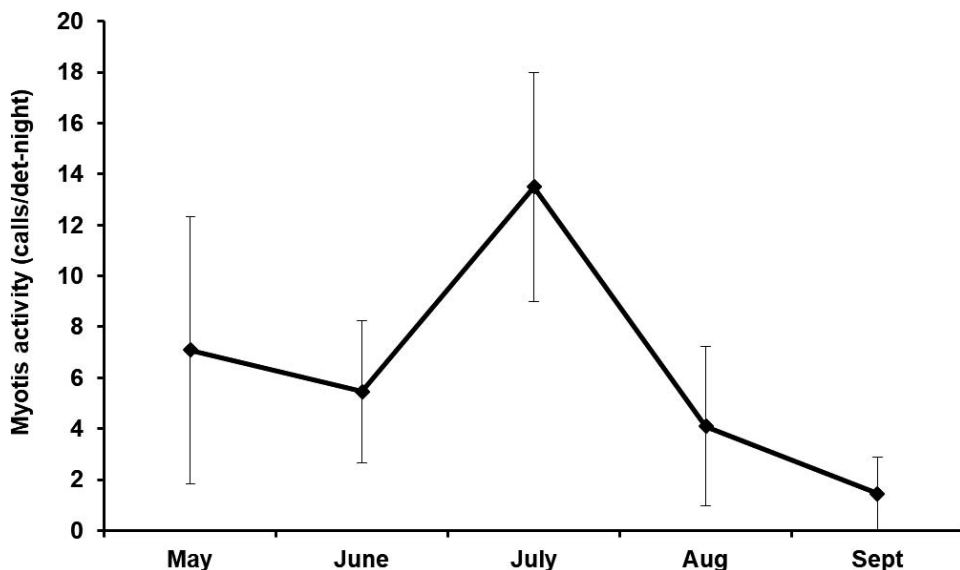


FIG. 1.—Call frequencies of *Myotis* bats (Mean $\pm$ SE) recorded by month in Laurel Ridge watershed in Robinson Forest, Kentucky, 2015–2018

to late July (Fig. 1). Activity of *Myotis* declined during August and dropped precipitously in September when *Myotis* bats had apparently left the forest ($F = 4.66$, $df = 4, 15$, $P = 0.013$). Despite WNS impacts across the region, overall summer activity of *Myotis* remained steady from 2015 to 2018 ($F = 0.77$, $df = 3, 16$, $P = 0.418$). Variation in call rates in 2017 and 2018 was high (Fig. 2).

Mist-netting efforts ($n = 103$ net-nights) resulted in the capture of 38 northern long-eared bats. Northern long-eared bats typically were captured in 2.6 m high nets over closed canopy, ridge top roads. A juvenile northern long-eared bat was captured over a ridge top road adjacent to the shelterwood harvest. Reproductive condition of captured northern long-eared bats represented all stages of the summer maternity season, including pregnant, lactating, and post-lactating females. The only other *Myotis* species captured was a single Indiana bat.

Ten of 16 northern long-eared bats (eight females, two males) were successfully tracked to 19 different day-roosts. Remaining bats were either unable to be located during daytime roost surveys or dropped their transmitter following release. Bats roosted in a five tree species: red maple (*Acer rubrum*, $n = 7$), scarlet oak (*Quercus coccinea*, $n = 7$), pitch pine (*Pinus rigida*, $n = 1$), black oak (*Q. velutina*, $n = 1$), tulip poplar (*Liriodendron tulipifera*, $n = 1$), and two unidentified snags (Table 1). Based upon random tree plots, red maple and scarlet oak were the most common species of cavity trees along ridge tops in the forest and occurred in various conditions from advanced decay with exfoliating bark, to declining trees with cavities, to live trees with smaller cavities. Evidence for roost selection by tree species was minimal ($\chi^2 = 9.11$, $df = 4$, $P = 0.059$). All roosts of northern long-eared bats were within 100 m of a ridge top road (Fig. 3), suggesting this bat chose roosts from among the most common dead and live tree species in the vicinity of these corridors. Northern long-eared bats roosted in both live (39%; $n = 7$) and dead trees (61%; $n = 11$).

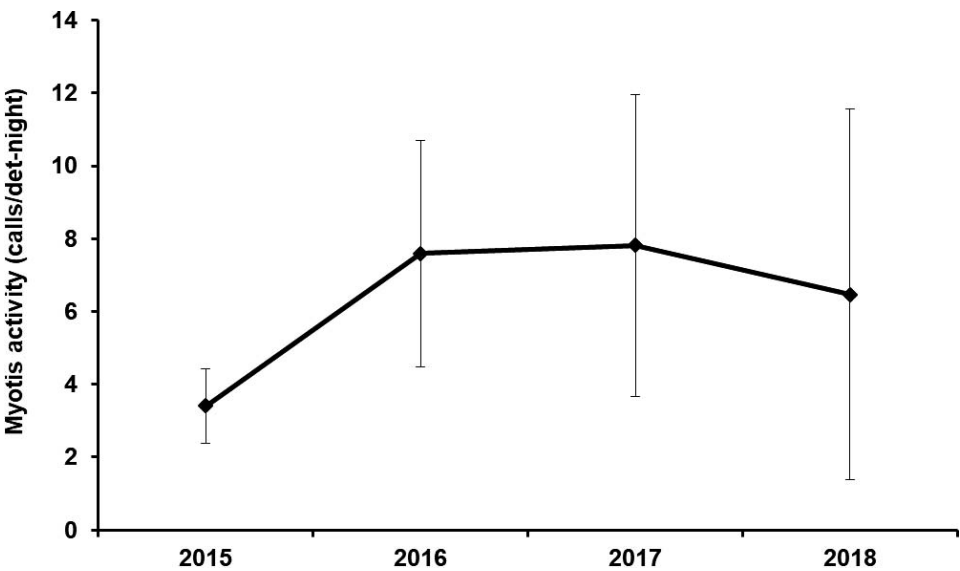


FIG. 2.—Call frequencies of *Myotis* bats (Mean $\pm$ SE) recorded across years in Laurel Ridge watershed in Robinson Forest, Kentucky, 2015–2018

In early May, before pregnancy was detected in captured bats, adult females often roosted solitarily in small cavities (85.7%) often large enough for only a single individual, in small diameter ($22.3\text{ cm} \pm 6.49\text{ SE}$) live and dead trees. During pregnancy through lactation, adult females clustered together beneath bark (75%), mostly in dead trees (87.5%) with larger diameters ($29.6\text{ cm} \pm 6.96\text{ SE}$). The difference between mean diameters was not significant ($t = 0.88$, $df = 14$, $P = 0.197$). Exit counts ($n = 26$) varied across the season (Fig. 4), with an overall average count of $5.15 \pm 1.29\text{ (SE)}$ bats. The maximum colony size recorded was 24

TABLE 1.—Number of cavity trees by species within random plots ($n = 18$) compared to the number of roost trees of northern long-eared bat by species at Robinson Forest, Kentucky, 2016–2017

Tree species	No. cavity trees	No. bat roosts
<i>Acer rubrum</i>	57	7
<i>Quercus coccinea</i>	30	7
<i>Oxydendrum arboreum</i>	21	0
<i>Q. montana</i>	16	0
<i>Q. velutina</i>	8	1
<i>Amelanchier arborea</i>	6	0
<i>Nyssa sylvatica</i>	5	0
<i>Q. alba</i>	4	0
<i>Acer saccharum</i>	4	0
<i>Liriodendron tulipifera</i>	4	1
<i>Magnolia acuminata</i>	3	0
<i>Q. rubra</i>	3	0
<i>Pinus rigida</i>	2	1

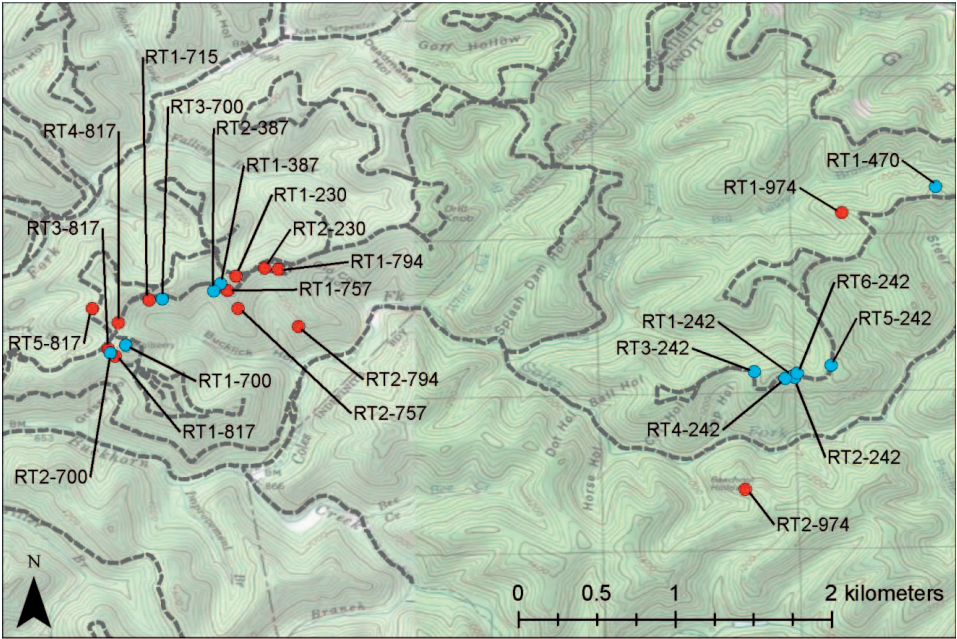


FIG. 3.—Location of roost trees of northern long-eared bats along forest roads at Robinson Forest, Kentucky, 2016–2017

bats in a dead red maple in an advanced stage of decay on 29 May 2016. The second largest colony size recorded was 21 bats in a dead black oak also in an advanced stage of decay on 10 June 2017. By late June, the number of bats emerging from colonies decreased although the type of roost chosen did not appear to change. Roost switching was minimal, with females

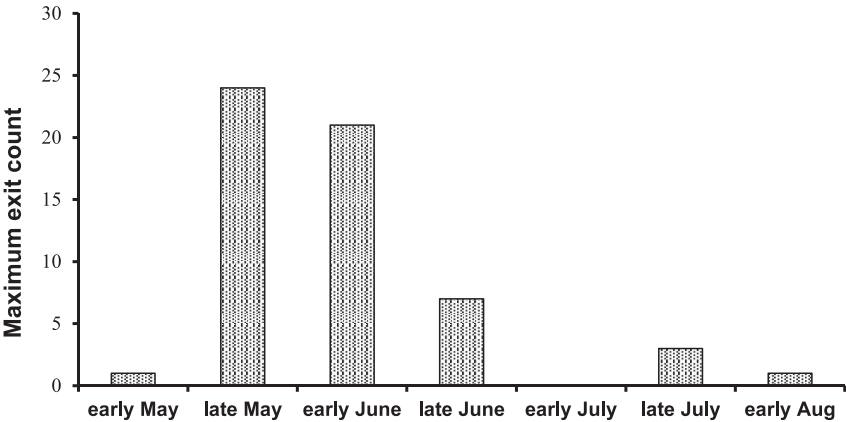


FIG. 4.—Maximum exit count from roost trees of northern long-eared bats by seasonal period at Robinson Forest, Kentucky, 2016–2017

often staying at the same site for several consecutive days. Exit counts during the post-volancy period were mostly of one or two individuals.

DISCUSSION

Historically, the northern long-eared bat was a common species across forested landscapes of eastern North America. Since arrival of WNS, summer populations have significantly declined across the geographic range of the species (Francl *et al.*, 2012; Reynolds *et al.*, 2016; Thalken *et al.*, 2018a). Scattered populations remain (Dobony and Johnson, 2018; Frank *et al.*, 2019), suggesting northern long-eared bats in some locations have persisted across multiple seasons of potential exposure to WNS (De La Cruz *et al.*, 2018). Should numbers of northern long-eared bats continue to decline from WNS across the distribution, surviving populations, including those within the Central Appalachian region, may be important to future population growth of these bats and re-occupancy of eastern North American forests. We successfully captured northern long-eared bats of all ages and reproductive classes in mist nets on Robinson Forest, and commonly detected *Myotis* calls in acoustic surveys in refute of our null hypothesis of low call detection rates. Further, levels of *Myotis* activity remained steady over the 4 y of sampling. Given that northern long-eared bats represented 97% of the *Myotis* bats captured in mist nets, the yearly patterns in total activity observed across all *Myotis*, likely paralleled the relative abundance of northern long-eared bats on the forest. Collectively, our data indicate a sustaining summer population of northern long-eared bats in an actively managed eastern Kentucky forest.

Not only have northern long-eared bats been shown to display strong genetic relatedness among local populations, but these bats also exhibit high rates of gene flow across regions (Johnson *et al.*, 2014). Given the extent of surface mining surrounding the study area, we suggest northern long-eared bats inhabiting Robinson Forest have limited interaction during summer months with other populations of this species across the region. Their departure from the forest in early autumn, though, as evidenced by few captures and sharp declines in number of echolocation calls recorded, suggests these bats leave the forest and migrate to swarming sites and winter hibernacula where mating and hibernating with individuals summering elsewhere likely occur. Presently, the hibernacula used by summer populations of northern long-eared bats on Robinson Forest remain unknown.

Northern long-eared bats showed variation in summer colony size, consistent with roosting patterns in other studies that demonstrate fission-fusion behaviors in these bats (Garroway and Broders, 2007; Johnson *et al.*, 2012; Silvis *et al.*, 2014; 2015a). Prior to WNS, Lacki and Schwierjohann (2001) observed seasonal patterns in colony size in populations of northern long-eared bats on the Daniel Boone National Forest, Kentucky, with largest exit counts from roosts occurring during pregnancy and subsequently declining through lactation and the weaning of young. Following exposure to WNS, populations of northern long-eared bats during spring staging in Mammoth Cave National Park, Kentucky, chose roosts in trees most likely to facilitate daytime torpor (Thalken and Lacki, 2018). Colonies were extremely small, averaging <4 bats/roost. Colonies of northern long-eared bats in our study also were small, with no exit count of 25 or more bats recorded and a seasonal average of <6 bats/roost. Prior to WNS, larger summer maternity colonies of northern long-eared bats were frequently reported (*e.g.*, Foster and Kurta, 1999 [n = 60]; Menzel *et al.*, 2002 [n = 65]; Owen *et al.*, 2002 [n = 88]; Perry and Thill, 2007 [n = 51]; Whitaker and Mumford 2009 [n = 100]), including in Kentucky (Lacki and Schwierjohann, 2001 [n = 65]; Lacki *et al.*, 2009b [n = 56]). Other studies on northern long-eared bats in regions following WNS exposure have reported large colonies persisting, at least in the short-term, in Tennessee

(Lereculeur, 2013 [n = 116]), Indiana (Badin, 2014 [n = 44]) and Kentucky (Silvis *et al.*, 2015b [n = 121]). Therefore, whether summer maternity colonies of northern long-eared bats in the WNS era eventually become smaller in size on average than historically observed remains unclear. Given that *Myotis* species rely on social thermoregulation during the summer maternity season (Webber and Willis, 2018), the potential effect, if any, of decreased group size on health and survivorship of juvenile northern long-eared bats warrants further study.

Although preference of tree species for roosting sites was only marginally significant ($P = 0.059$) on Robinson Forest, northern long-eared bats relied predominantly on two species of trees for roosting, red maple and scarlet oak. Evidence for use of only one or two tree species for roosting by local populations of northern long-eared bats has been observed elsewhere, with the tree species chosen varying across sites (Lacki and Schwierjohann, 2001; Ford *et al.*, 2006; Perry and Thill, 2007; Silvis *et al.*, 2012). These patterns are consistent with the northern long-eared bat's known flexibility in roost choice, wide range of tree species used and ability to adapt to the local mix of trees present (Silvis *et al.*, 2016b). Measured diameters of roost trees of northern long-eared bats in this study increased approximately 33% from early spring to pregnancy through lactation, but the increase was not significant ($P = 0.197$). Mean diameters of trees used on Robinson Forest by northern long-eared bats were within the wide range of diameters reported across the distribution of the species (Lacki *et al.*, 2009a; Silvis *et al.*, 2016b).

Robinson Forest is a largely intact, experimental forest possessing a network of old logging roads throughout the main block. We believe the extensive system of old, maintained harvest roads provided suitable habitat for northern long-eared bats on Robinson Forest. All captures of northern long-eared bats occurred on forest roads and not in other habitats sampled, and these bats roosted exclusively in trees along ridge tops within 100 m of these roads. Studies in central Kentucky have also demonstrated northern long-eared bats to select trees along or near forested ridge tops as roosting sites (Thalken *et al.*, 2018b; Thalken and Lacki, 2018), and across their distribution northern long-eared bats have shown a tendency toward use of higher elevation, upland forests for roosting (Lacki and Schwierjohann, 2001; Jung *et al.*, 2004; Johnson *et al.*, 2009; Silvis *et al.*, 2012, 2016b). Not only is it highly likely that forest roads on Robinson Forest served as flyways for northern long-eared bats to navigate across the entire forest, but also as feeding corridors, where this species could glean prey off the surface of tunneled vegetation, and rock and ground surfaces (Faure *et al.*, 1993; Broders *et al.*, 2004). Use of roads as corridors by bats within managed forests has been observed elsewhere (Hein *et al.*, 2009), although the effects of roads on bats in heavily fragmented and human-altered habitats is perceived to be negative (Altringham and Kerth, 2016).

Loss of intact forest and potential roosting sites due to logging activities is believed to locally, but temporarily, impact habitat of northern long-eared bats (Patriquin and Barclay, 2003; Silvis *et al.*, 2015a). As harvested stands regenerate, however, the heavily compacted skid trails and harvest roads do not re-grow trees. These pathways subsequently provide corridors of unobstructed flight space enclosed within closed-canopy forest. These strips of habitat can serve as semi-permanent flyways within managed forests supporting flight activity of several species of bats, especially *Myotis* (O'Keefe *et al.*, 2013; Silvis *et al.*, 2016a; Ketzler *et al.*, 2018; Caldwell *et al.*, 2019). When planning harvests in eastern North American forests, we encourage maintaining unpaved roads, especially along ridge tops, in managed forests to promote long-term roosting habitat of northern long-eared bats. Roads should be designed to have increased canopy closure as surrounding habitat develops post-harvest. Maximizing connections among roads located along ridge tops is also encouraged, as this can lead to a

flyway matrix throughout the forest landscape. Such a matrix might further enhance feeding activity and commuting behavior of forest-dwelling bats within these corridors. If possible, any snag or tree with cavities next to a ridge top road should be surveyed for occupancy by northern long-eared bats before being felled, as our data indicate these trees are likely to be potential roosting habitat.

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